RV Convergence

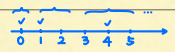
s　完全收敛　$\lim_{n\to\infty} X(n,s) = X(s)$　$\forall \varepsilon>0,\ \exists N$ when $n>N$ s.t. $|X_n - X| < \varepsilon$

a.s.　a.s. 收敛　$Pr\{\lim_{n\to\infty} X(n,s) = X(s),\ \forall s\} = 1$

P　依概率收敛　$\lim_{n\to\infty} Pr\{|X(n,s) - X(s)| > \varepsilon,\ \forall s\} = 0,\ \forall \varepsilon$

MS　均方收敛　$E[(X(n,s) - X(s))^2] = 0$

d　依分布收敛　$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$



完全收敛 $\Rightarrow$ a.s. 收敛 $\overset{\not\Leftarrow}{\Rightarrow}$ 依概率收敛 $\Rightarrow$ 依分布收敛

均方收敛 $\overset{\Rightarrow}{\not\Leftarrow}$ $e^{-n(ns-1)}$

Markov 不等式

若 X 取值非负　$P(X \geq a) \leq \frac{E(X)}{a}$

$E(X) = \int_0^\infty x f_X(x)dx \geq \int_a^\infty x f_X(x)dx \geq \int_a^\infty a f_X(x)dx = aP(X \geq a)$

Chebyshev 不等式　$P\{|X - E(X)| > a\} \leq \frac{\sigma_X^2}{a^2}$　Markov 不等式取 X 为 $|X-E(X)|$ 即可

Chernoff 界　$Pr(X \geq a) \leq \min_{s\geq 0}\{e^{-sa} M_X(s)\}$

大数定理：均值逼近期望

WLLS　若 μ_X 有限，则 $\bar{X}$ 依概率收敛于 μ_X

即 $\lim_{n\to\infty} P\{|\bar{X} - \mu_X| < \varepsilon\} = 1$　Chebyshev 不等式取极限即可

SLLS　若 μ_X、σ_X^2 都有限，则 $\bar{X}$ a.s. 收敛于 μ_X

即 $Pr\{\lim_{n\to\infty} \bar{X} = \mu_X,\ \forall s\} = 1$

中心极限定理 CLT：均值正态分布

$$\lim_{n\to\infty} P\left(\frac{\bar{X} - \mu_X}{\sigma_X/\sqrt{n}} \leq z\right) = \Phi(z)$$

统计中的3种分布

χ^2 分布：标准高斯分布的平方和

$x_i \sim N(0,1)$，　$\chi_N^2 = \sum_{i=1}^{N} x_i^2$

统计量 $E(\chi_N^2) = N$，$Var(\chi_N^2) = 2N$

t 分布：　$T = \frac{X}{\sqrt{Y/n}}$　　$X \sim N(0,1)$，$Y \sim \chi^2(n)$

F 分布：　$F = \frac{X/m}{Y/n}$　　$X \sim \chi^2(m)$，$Y \sim \chi^2(n)$

$\bar{X}$ 和 S^2

若 $x_1 \cdots x_n \sim N(\mu, \sigma^2)$，则 $\bar{X}$，S^2 有重要性质

样本均值 $\bar{X} = \frac{1}{N}\sum x_n$

$\bar{X}$ 是 μ 的无偏估计 $E(\bar{X}) = \mu$

$\bar{X}$ 是 μ 的有效估计，即 MVU 估计

$Var(\bar{X}) = \frac{\sigma^2}{n}$　而正态分布的 CRLB 恰为 $\frac{\sigma^2}{n}$

$\bar{X}$ 也服从正态分布 $N(\mu, \frac{\sigma^2}{n})$

渐近时任意分布的 $\bar{X}$ 都有 $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

样本方差 $S^2 = \frac{1}{N-1}\sum(x_n - \bar{x})^2$

S^2 是 σ^2 的无偏估计

$E(S^2) = \sigma^2$

$$
\begin{aligned}
E(S^2) &= \frac{1}{N-1} E[\sum(x_n - \bar{x})^2] \\
&= \frac{1}{N-1} E[\sum(x_n - \mu + \mu - \bar{x})^2] \\
&= \frac{1}{N-1} E[\sum(x_n - \mu)^2 + (\mu - \bar{x})^2 + 2(x_n - \mu)(\mu - \bar{x})] \\
&= \frac{1}{N-1}\{\sum E[(x_n - \mu)^2] + \sum E[(\mu - \bar{x})^2] + 2\sum E[(x_n - \mu)(\mu - \bar{x})]\} \\
&= \frac{1}{N-1}\{N\sigma^2 + N \cdot \frac{\sigma^2}{N} - 2\sigma^2\} \\
&= \sigma^2
\end{aligned}
$$

$Var(\bar{x}) = \frac{\sigma^2}{N}$

$$
\begin{aligned}
&E[(x_n - \mu)(\bar{x} - \mu)] \\
&= \frac{1}{N} E[(x_n - \mu)(\sum x_i - \mu)] \\
&= \frac{1}{N} \sum_{i=1}^{N} E[(x_n - \mu)(x_i - \mu)] \\
&= \frac{1}{N}[\cdots + 0 + \sigma^2 + 0 + \cdots]
\end{aligned}
$$

S^2 服从 χ^2 分布　　$\frac{(n-1)}{\sigma^2} S^2 \sim \chi^2(n-1)$